

AMD.L.I-4.pdf



Anónimo



Automatas y Matematicas Discretas



1º Grado en Ingeniería Informática del Software



Escuela de Ingeniería Informática
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Lexical Analysis - AMD.L.I-4 TEST
Theory of Automata and Discrete Mathematics
School of Computer Science
University of Oviedo
07-03-2023

Download LA_G4.zip and unzip it.

Edit the file examSpec.lex to do the following exercises:

1. Exercise 1 (3 points)

Add two new rules for an operator and a reserved word as follows as follows:

- Decrement. The operator code will be -- and the output message should be
SCANNER:: found Operator DECREMENT
- Class. The reserved word to match will be class and the output message should be
SCANNER:: found Reserved Word CLASS

2. Exercise 2 (3.5 points)

Write a new rule so your scanner recognizes exam codes. An exam code consists of 2 sections separated by a dot (.). The first section consists of a digit in {1, 3, 7}, followed by an odd number of lowercase letters (at least 1). The second section consists of a sequence of 1 or more digits in {7, 8, 9} that ends in 9, followed by 3 or 4 uppercase letters. The output message should be SCANNER:: found EXAM CODE (the exam code)

Correct examples: 7a.9ABC, 3enl.8978989MMMM, 1barww.999QWER, 3mmm.79CBA

Incorrect examples: a.9ABC, 3enl.8978989MMMMMA, 1barww-999QWER, 3mmm.19CBA

3. Exercise 3 (3.5 points)

Write a new rule so your scanner recognizes lists of exam codes. A list of exam codes consists of a sequence of 1 or more exam codes separated by commas (,) and enclosed in parentheses (). Note that **the empty list () is not allowed**, and there cannot be empty spaces in the list. The output message should be SCANNER:: found LIST OF EXAM CODES (the list of exam codes)

Correct examples: (7a.9ABC), (7a.9ABC,1barww.999QWER,7a.9ABC), (7a.9ABC,3mmm.79CBA,3mmm.79CBA), (7a.9ABC,7a.9ABC,7a.9ABC,7a.9ABC)

Incorrect examples: (a), {7a.9ABC}, (7a.9ABC 7a.9ABC,7a.9ABC), ()

You can test your scanner generating it with the command `generateScanner.bat` and running it with the file `test_exam` as input with the command `java -classpath ; Yylex test_exam`

When done, upload your file `examSpec.lex` to the task in Campus Virtual.



do your thing

WUOLAH

Expected output

```
SCANNER:: found Operator DECREMENT
SCANNER:: found Reserved Word CLASS
SCANNER:: found EXAM CODE (7a.9ABC)
SCANNER:: found EXAM CODE (3enl.8978989MMMM)
SCANNER:: found EXAM CODE (1barww.999QWER)
SCANNER:: found EXAM CODE (3mmm.79CBA)
SCANNER:: LEXICAL ERROR (a.9ABC)
SCANNER:: LEXICAL ERROR (3enl.8978989MMMMMA)
SCANNER:: LEXICAL ERROR (1barww-999QWER)
SCANNER:: LEXICAL ERROR (3mmm.19CBA)
SCANNER:: found LIST OF EXAM CODES ((7a.9ABC))
SCANNER:: found LIST OF EXAM CODES ((7a.9ABC,1barww.999QWER,7a.9ABC))
SCANNER:: found LIST OF EXAM CODES ((7a.9ABC,3mmm.79CBA,3mmm.79CBA))
SCANNER:: found LIST OF EXAM CODES ((7a.9ABC,7a.9ABC,7a.9ABC,7a.9ABC))
SCANNER:: LEXICAL ERROR ((a))
SCANNER:: LEXICAL ERROR ({7a.9ABC})
SCANNER:: LEXICAL ERROR ((7a.9ABC 7a.9ABC,7a.9ABC))
SCANNER:: LEXICAL ERROR (())
```